

Ethanol Blending in Gasoline – Greece

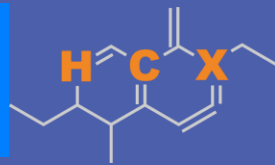
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**U.S. GRAINS &
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Ethanol Blending in Europe

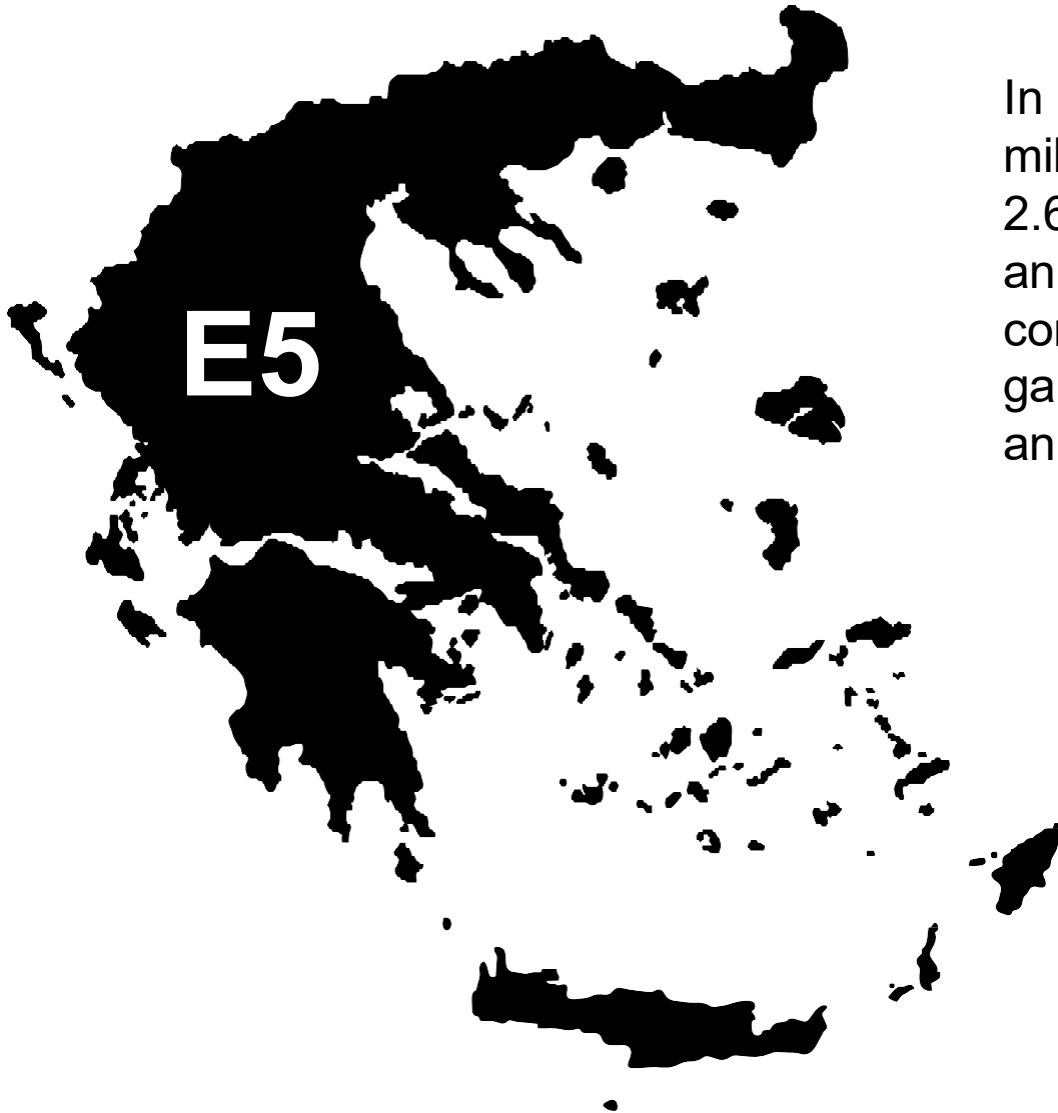
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Greece

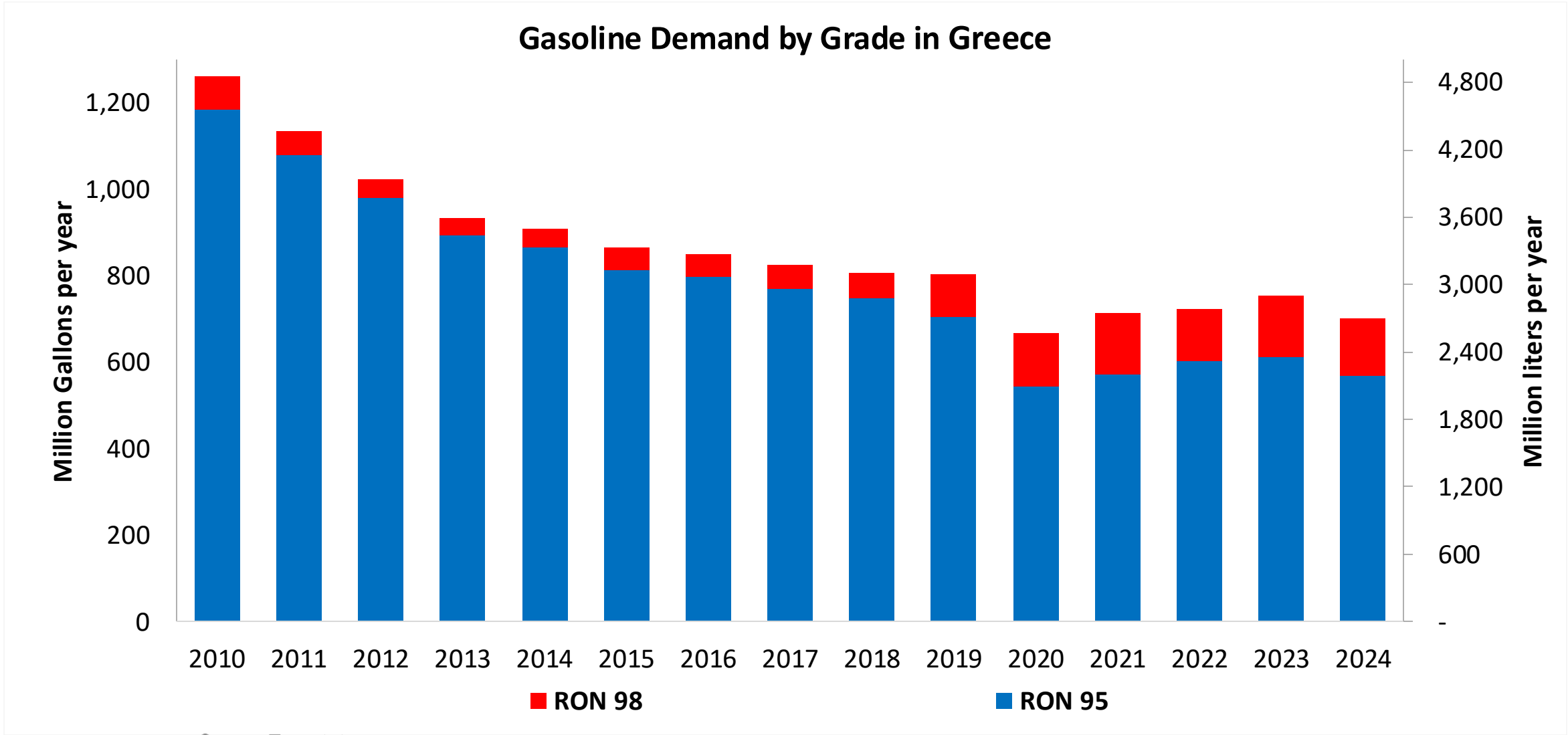


In 2024, gasoline consumption in Greece was 2,660 million liters (703 million gallons) and growth rate was -2.6%. Gasoline production and net imports were 2,811 and -151 million liters (743 and -40 million gallons) correspondingly. Greece complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

Ethanol consumption in Greece is determined by 3.3% bioethanol content mandate (by energy). Demand was 134 million liters (35 million gallons) representing 5.0% of gasoline demand. Ethanol is supplied by imports.

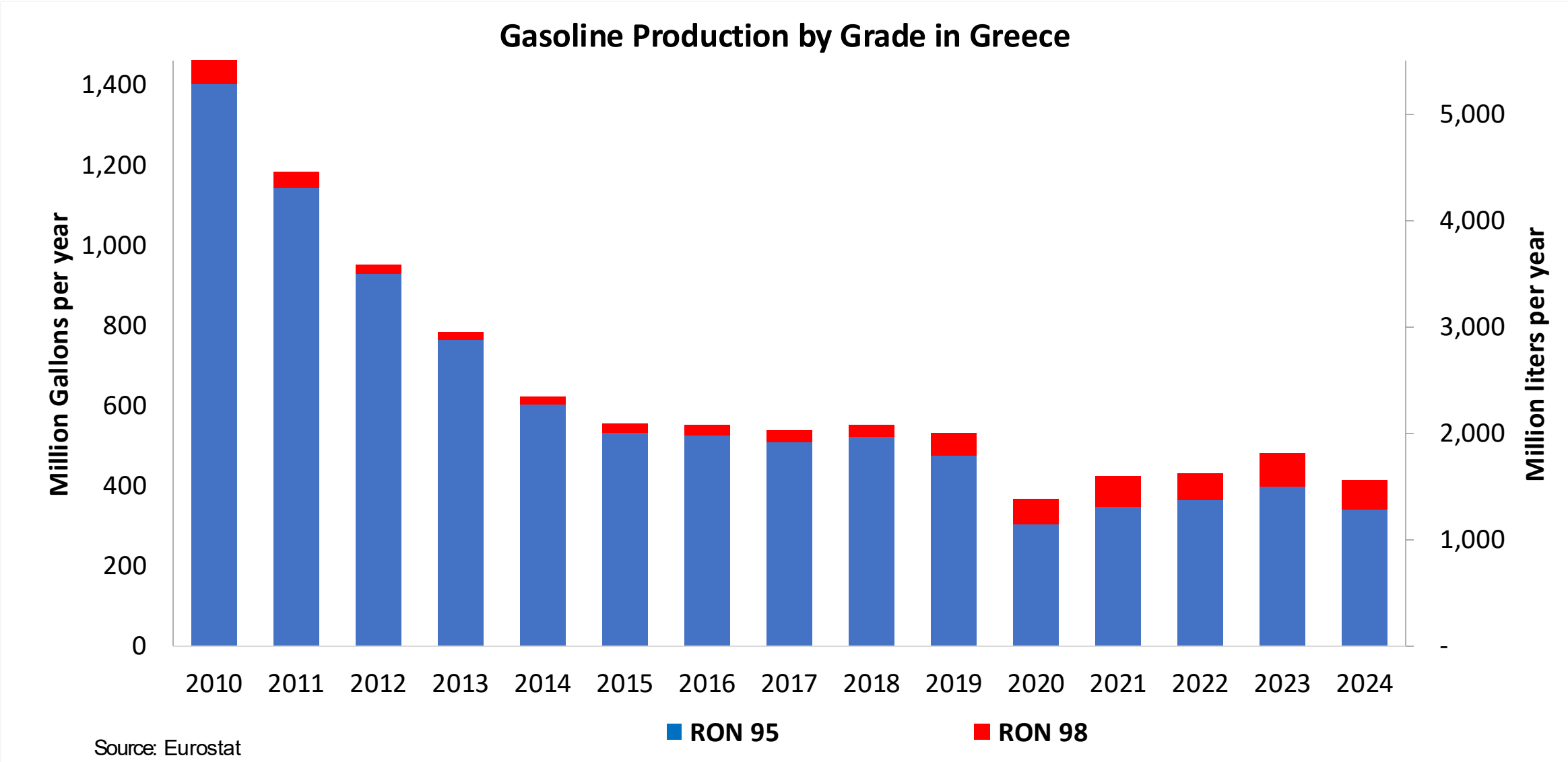
Source: Eurostat, European Commission, USDA

Gasoline Demand in Greece

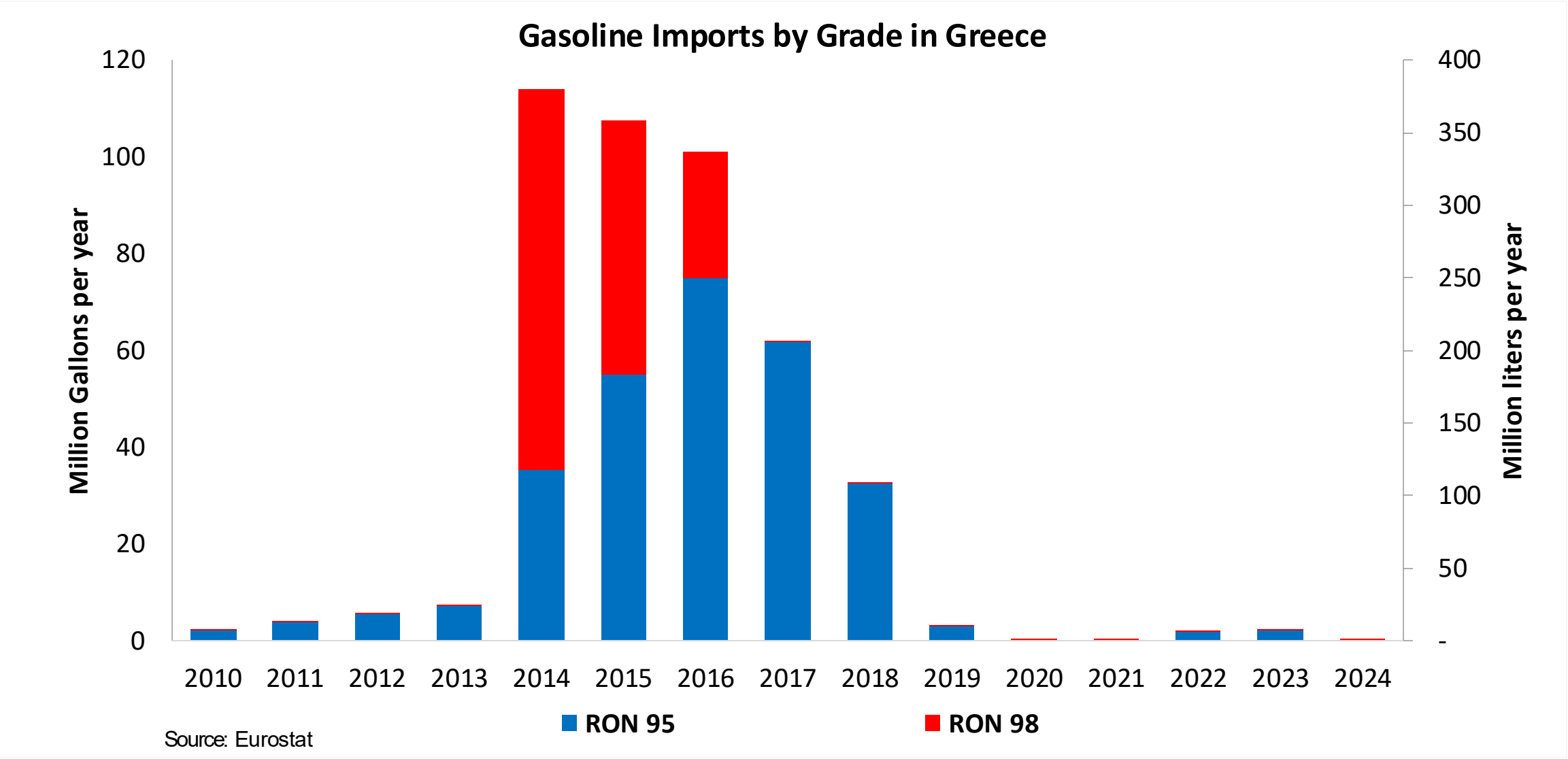


Source: Eurostat

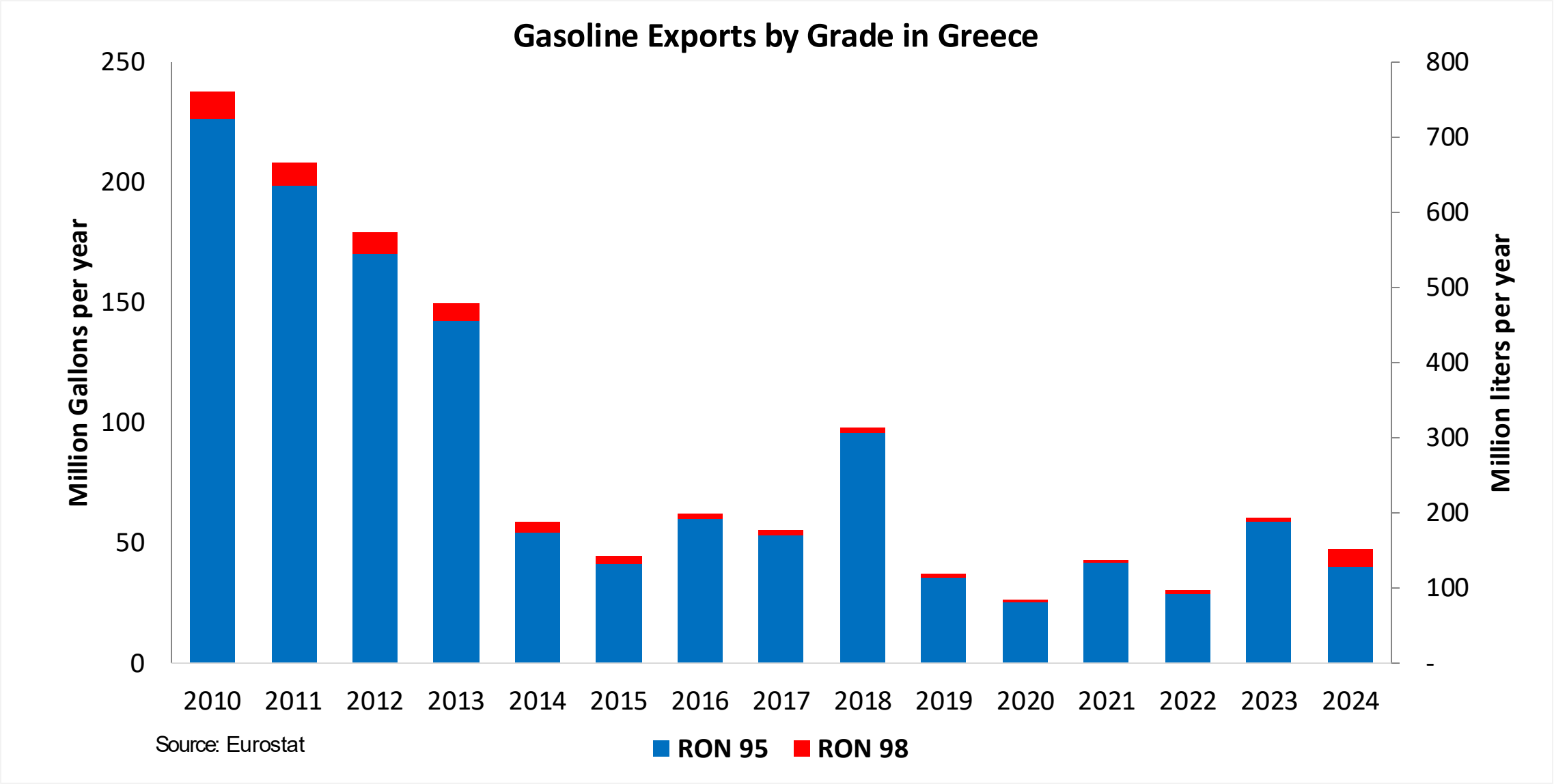
Gasoline Production in Greece



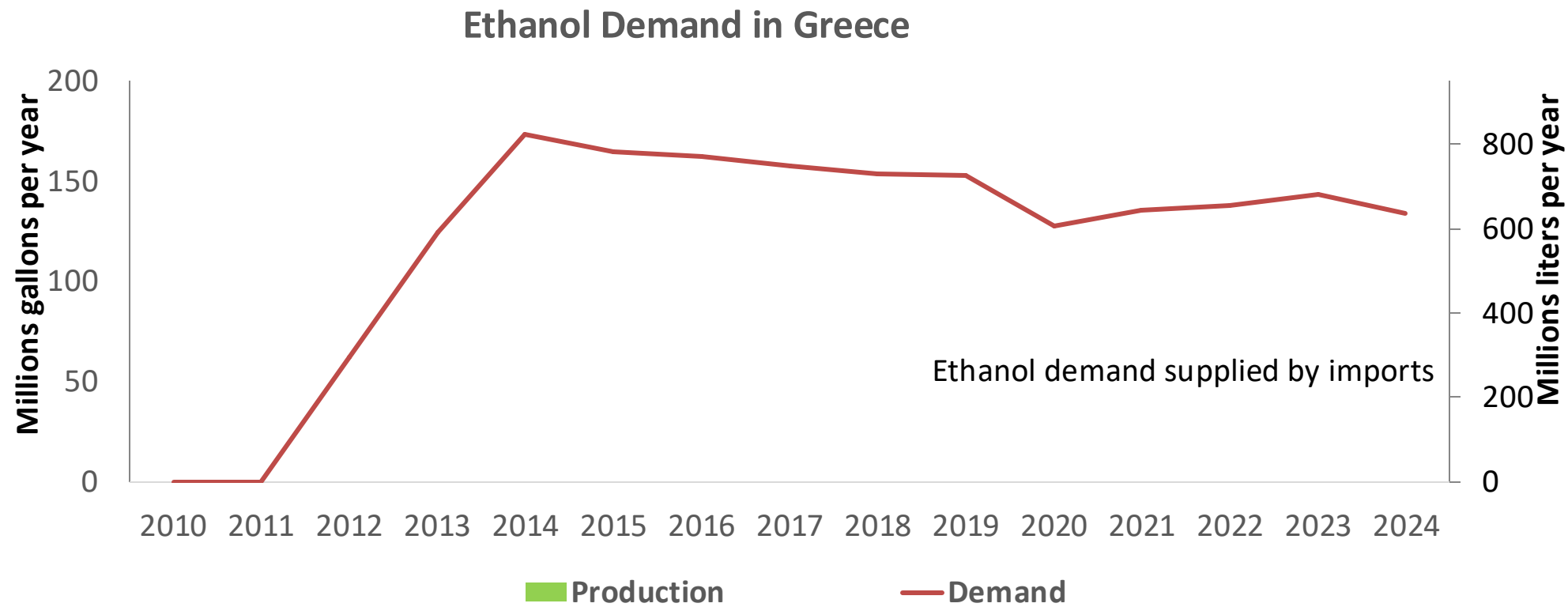
Gasoline Imports in Greece



Gasoline Exports in Greece



Ethanol Balance in Greece



Source: Eurostat, USDA – Fuel Mandates in EU

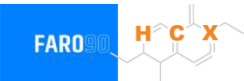
Gasoline Quality in Greece

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	3.3							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

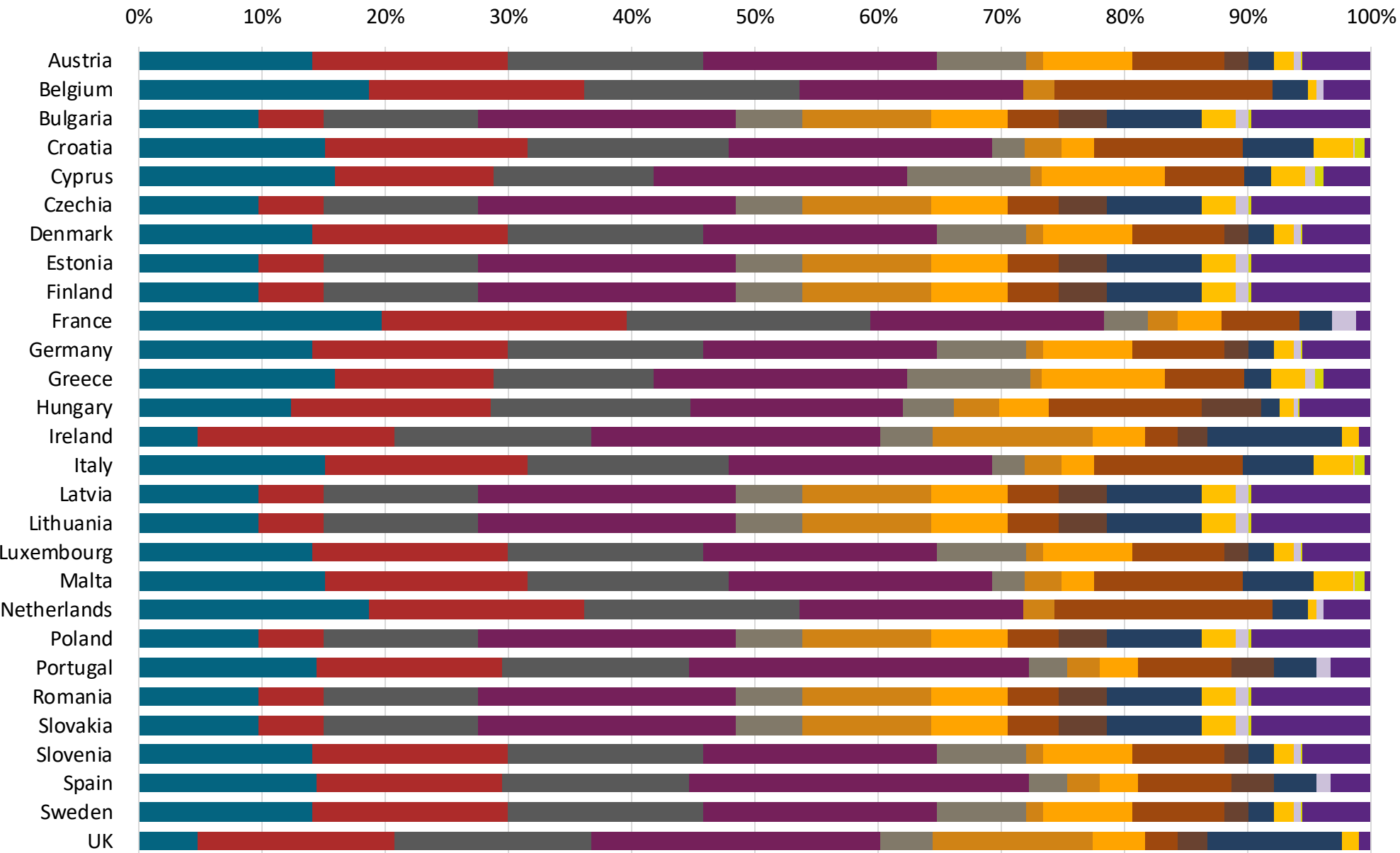
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



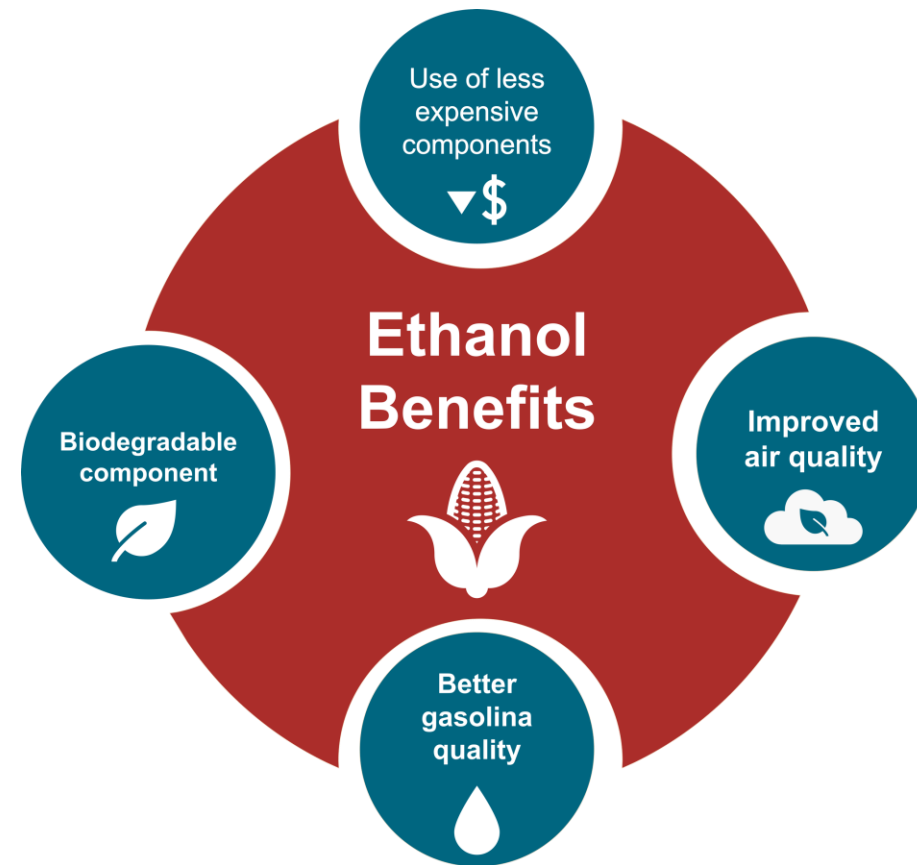
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

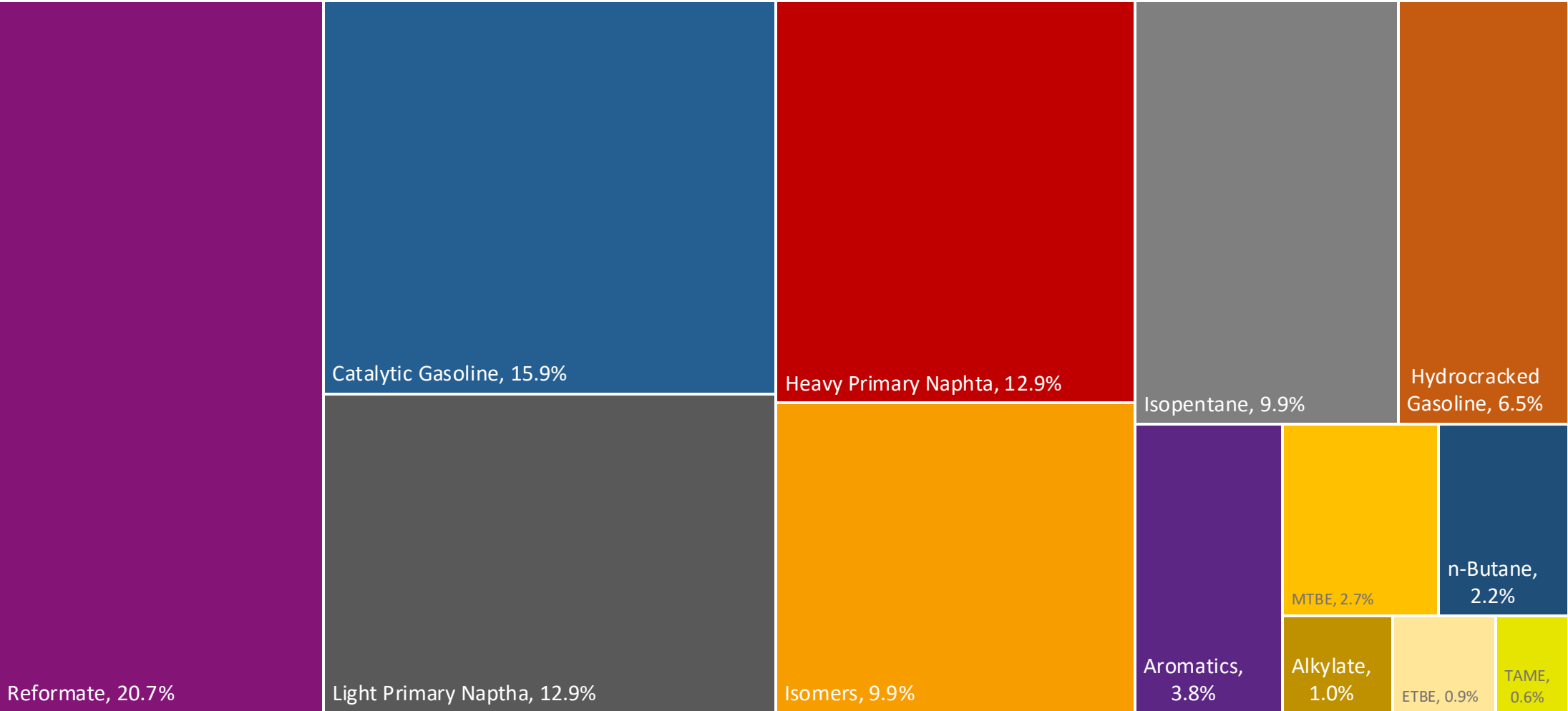
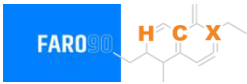
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

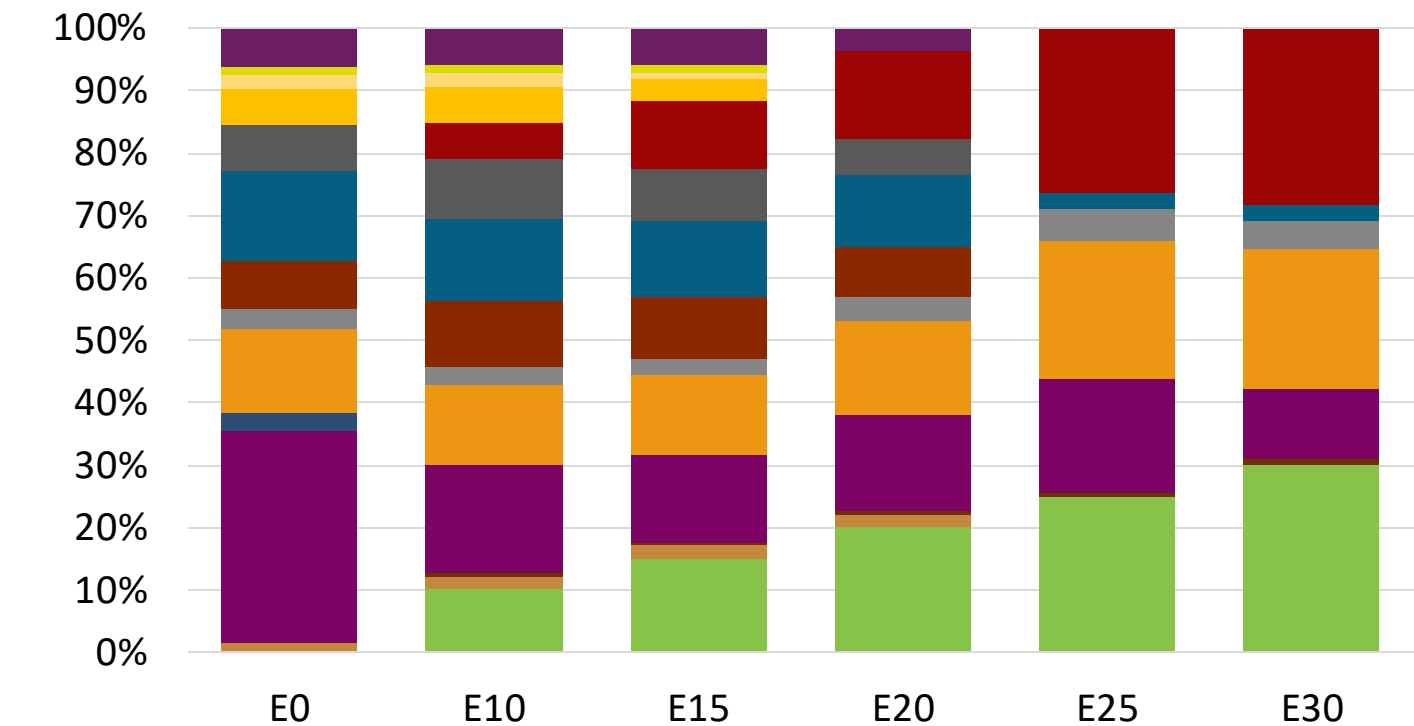
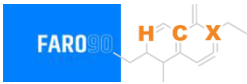
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Greece Available Blending Components



Greece – Gasoline 95 – Constant Octane

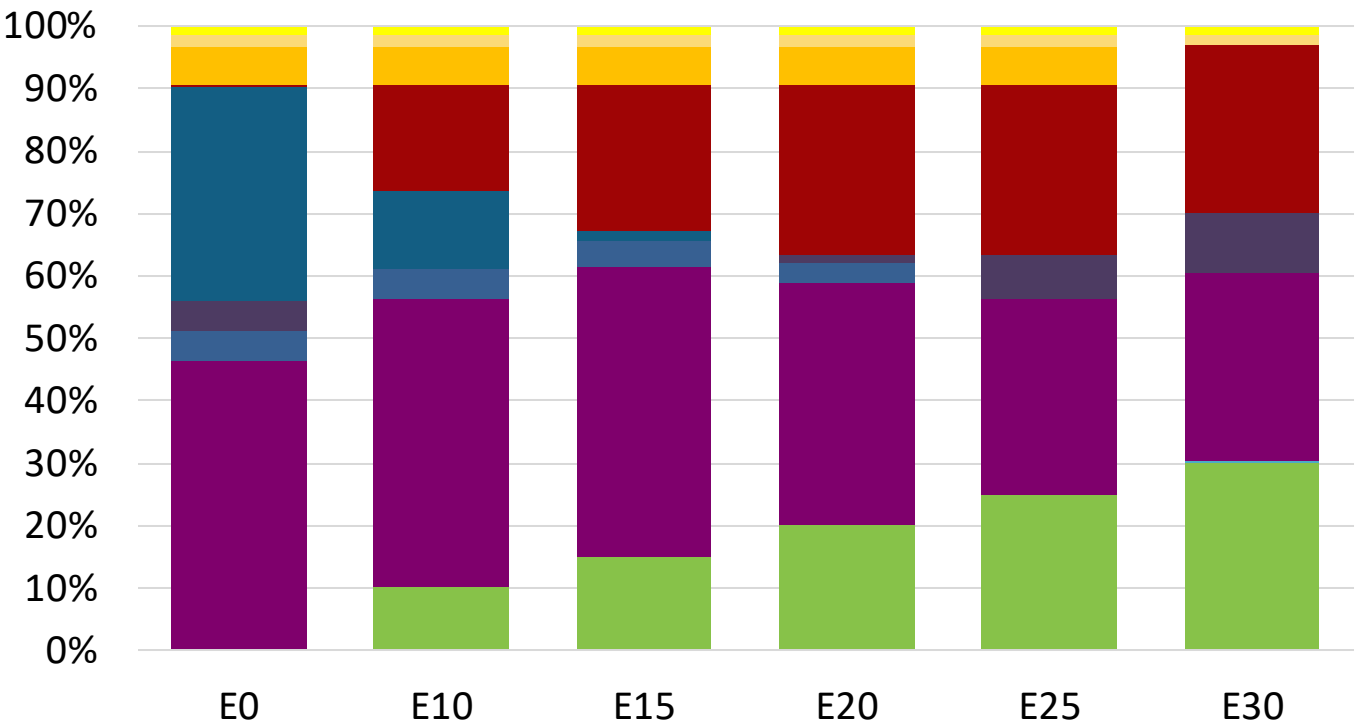


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US\$/gal)	\$ 2.258	\$ 2.135	\$ 2.069	\$ 2.002	\$ 1.881	\$ 1.845

Source: Faro90

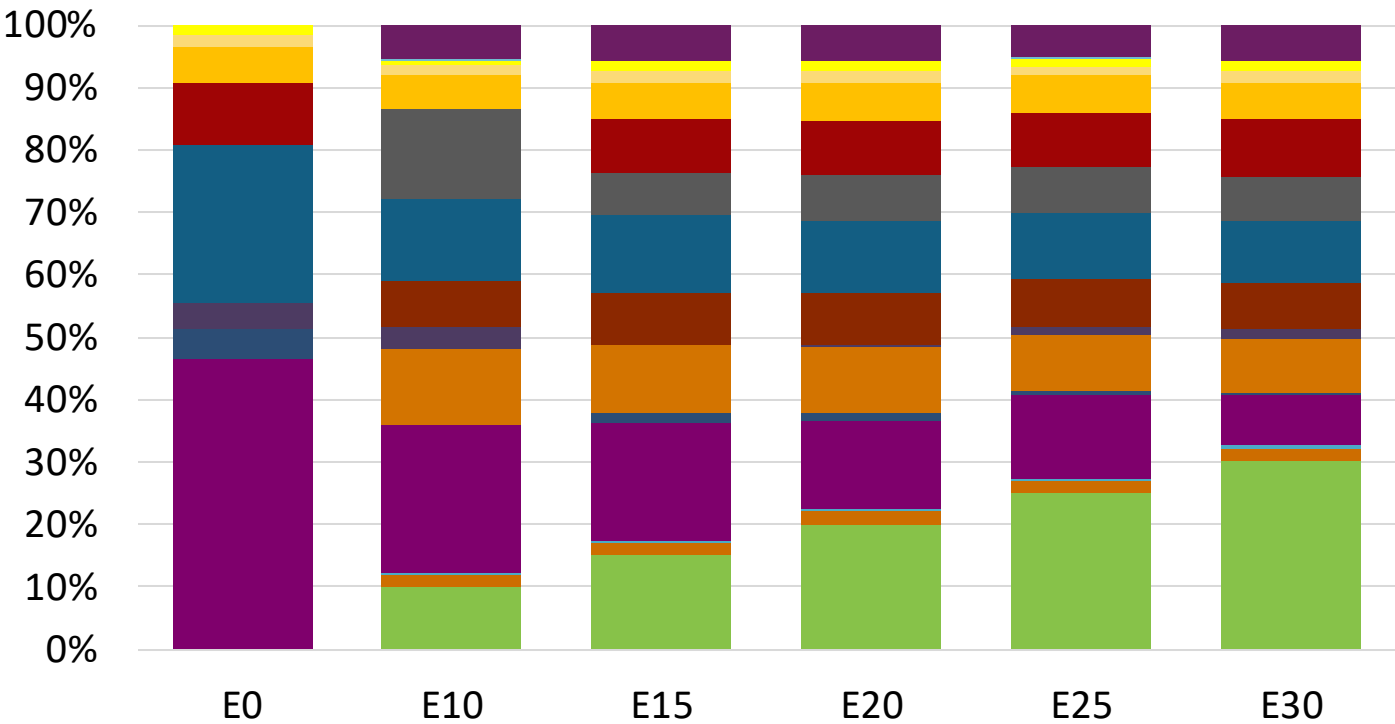
Greece - Gasoline 98 - Constant Octane



Average CIF 2024
Prices

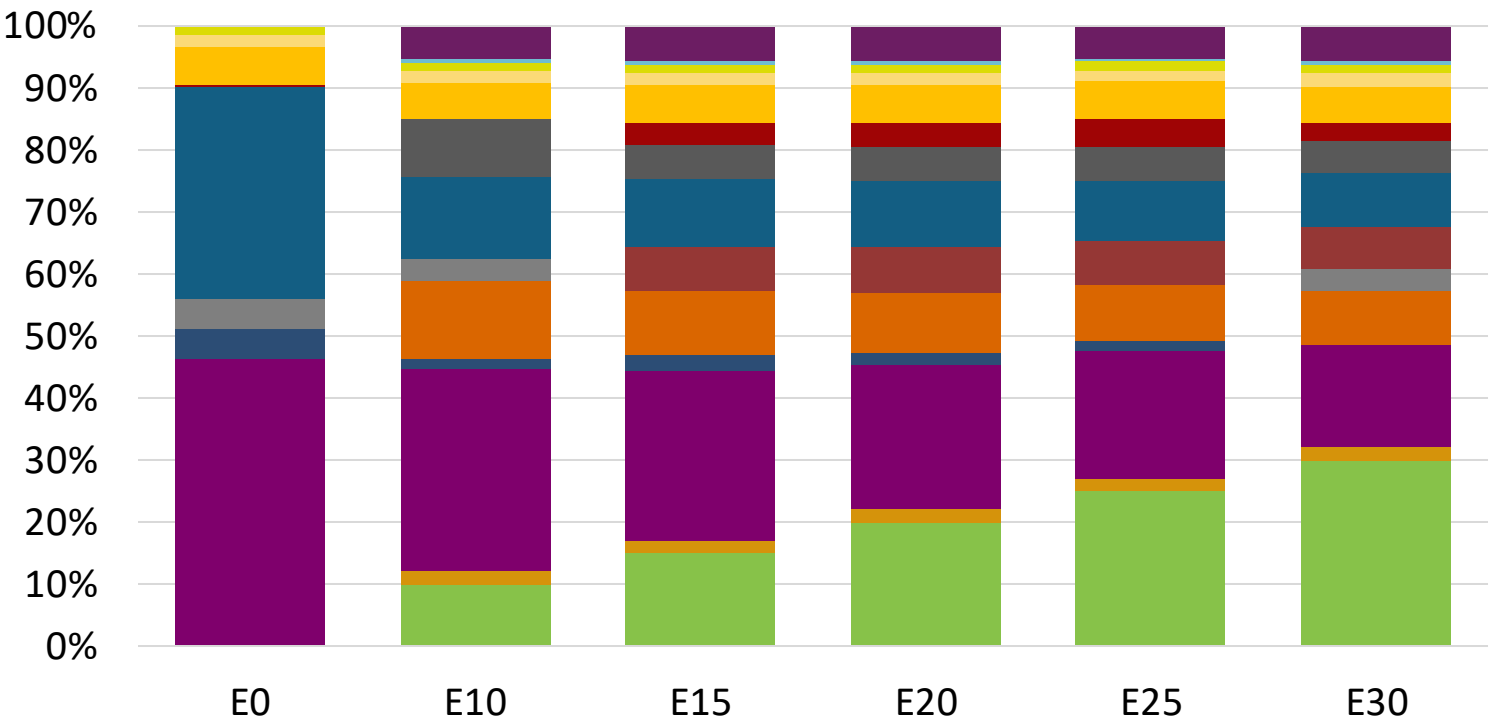
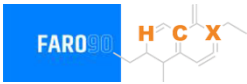
Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.330	\$ 2.204	\$ 2.132	\$ 2.057	\$ 1.980	\$ 1.924

Greece – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Greece – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.4	100.7	102.0	103.3	104.5
Price (US\$/gal)	\$ 2.330	\$ 2.330	\$ 2.330	\$ 2.330	\$ 2.330	\$ 2.330

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

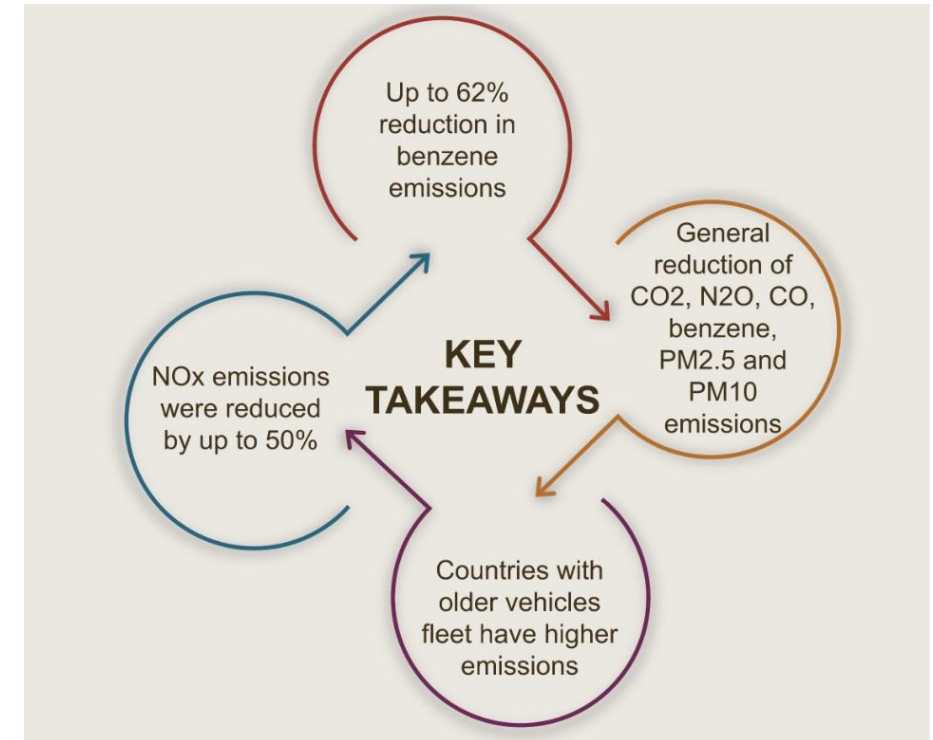
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

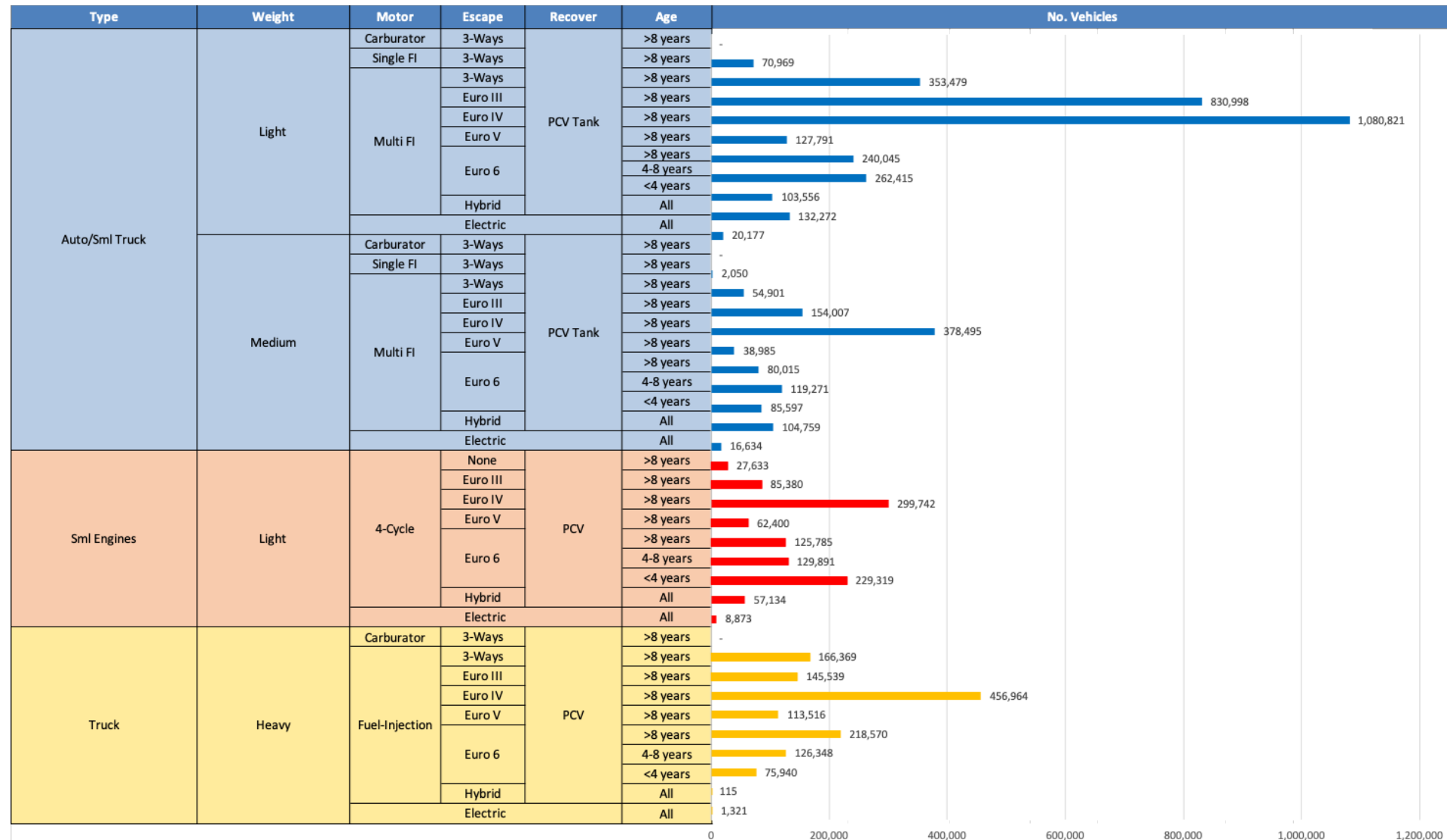
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Greece

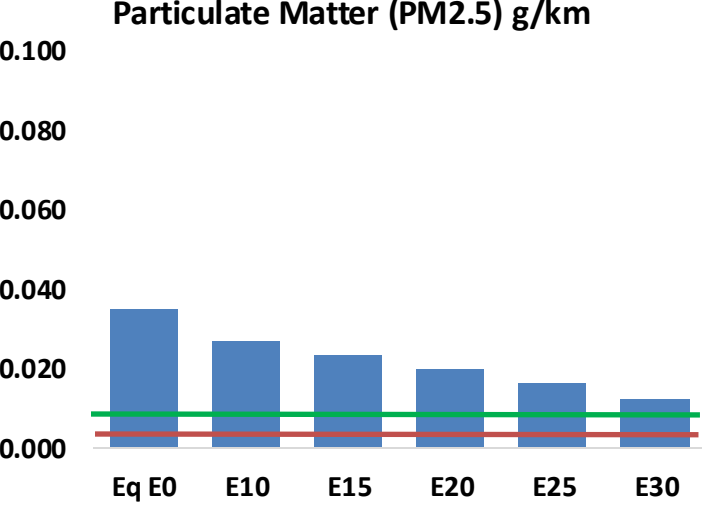
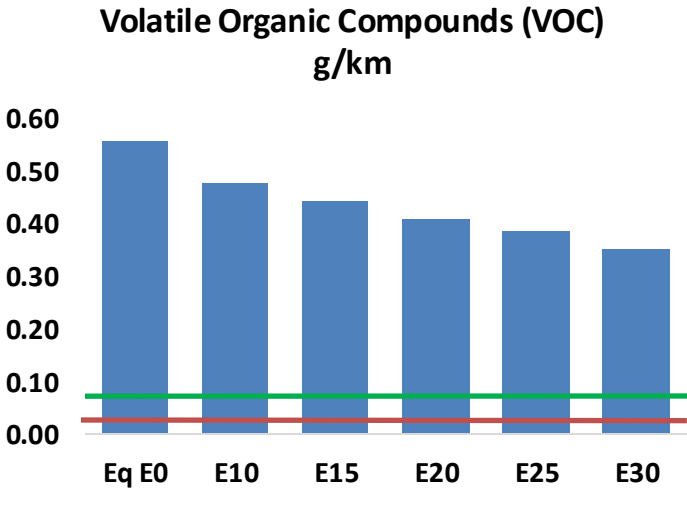
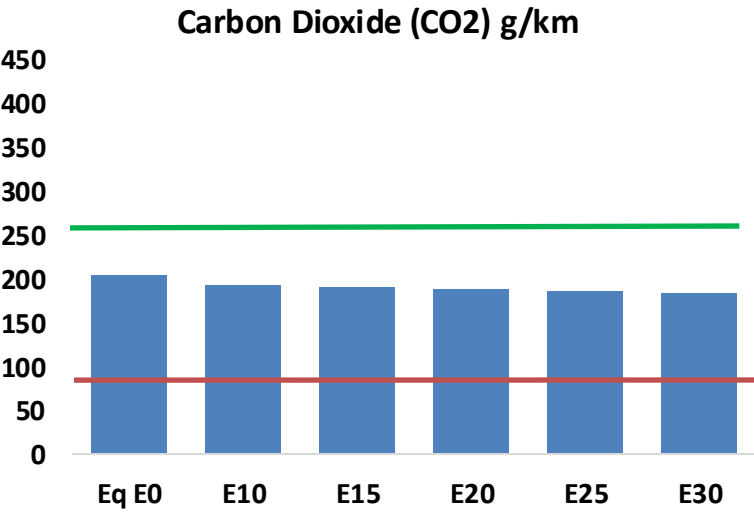
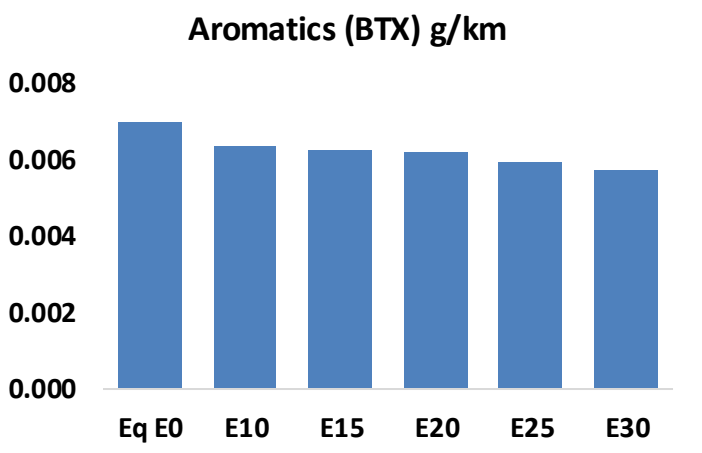
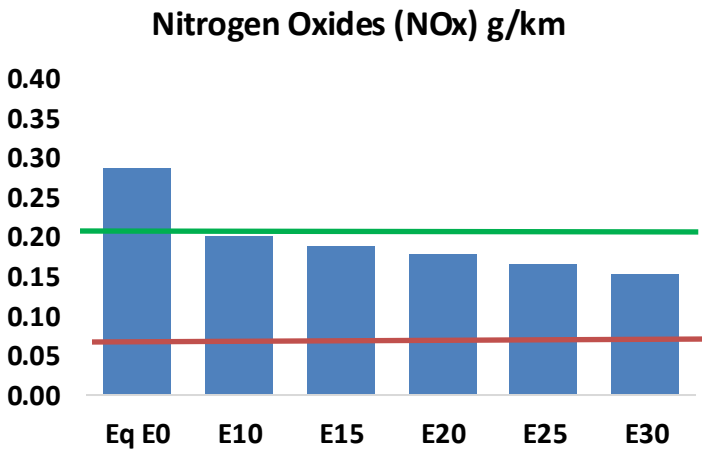
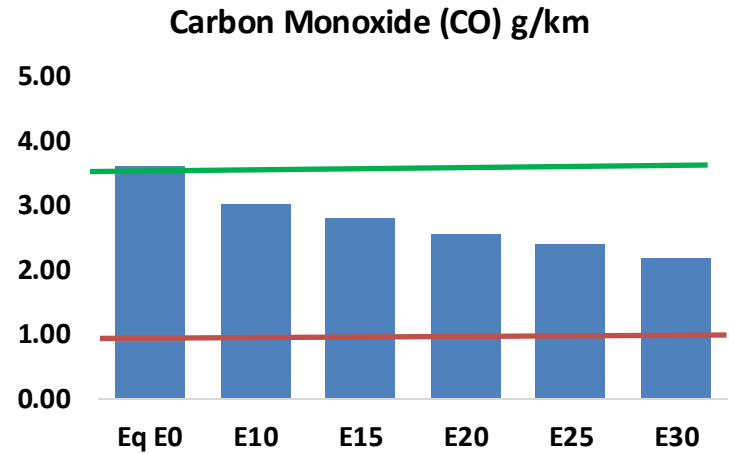
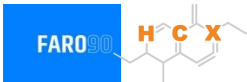


Vehicle Fleet: 6,588,079
Gasoline Vehicle Fleet: 4,489,264

Source: Eurostat, ACEA

Greece – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Greece – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	385,776	355,555	325,335	300,460	275,378	256,076	234,621	-16%	-29%
CO2	21,880,668	21,318,022	20,786,634	20,368,713	20,161,910	20,045,323	19,675,830	-5%	-8%
VOC	59,679	55,319	50,959	47,431	43,901	41,193	37,557	-15%	-26%
NOx	30,711	23,034	21,498	20,262	19,109	17,837	16,492	-30%	-38%
SOx	260	240	221	204	188	171	153	-15%	-28%
NH3	7,990	7,911	7,832	7,869	7,958	8,020	8,072	-2%	0%
N2O	301	299	298	299	305	309	312	-1%	2%
CH4	12,854	12,854	12,854	13,047	13,330	13,574	13,806	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	391	376	360	351	342	336	326	-8%	-13%
Acetaldehyde	1,097	1,259	1,847	2,812	3,831	4,378	5,173	68%	249%
Formaldehyde	3,922	3,813	4,445	5,220	5,469	6,050	6,586	13%	39%
Benzene	749	719	682	671	665	636	616	-9%	-11%
PM2.5	1,379	1,225	1,070	929	789	640	485	-22%	-43%
PM10	2,338	2,076	1,814	1,575	1,337	1,085	822	-22%	-43%

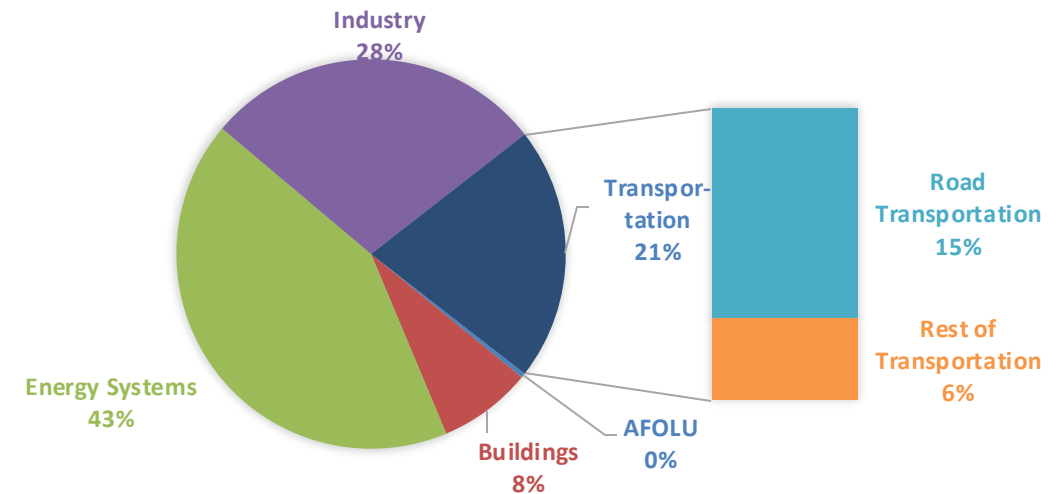
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

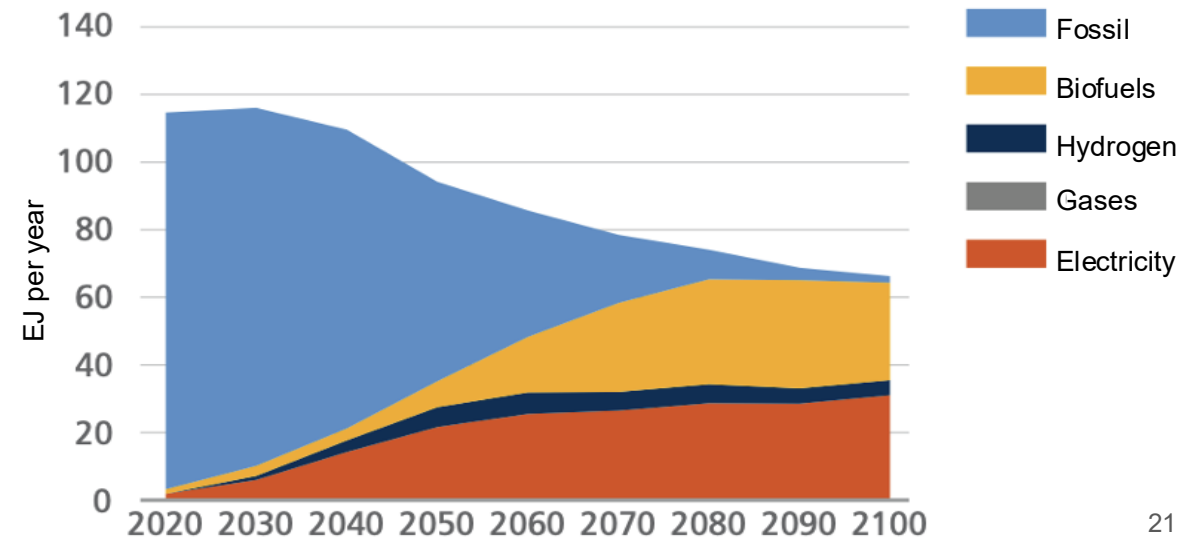
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

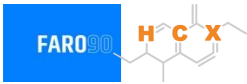
Current GHG Global Emissions Distribution



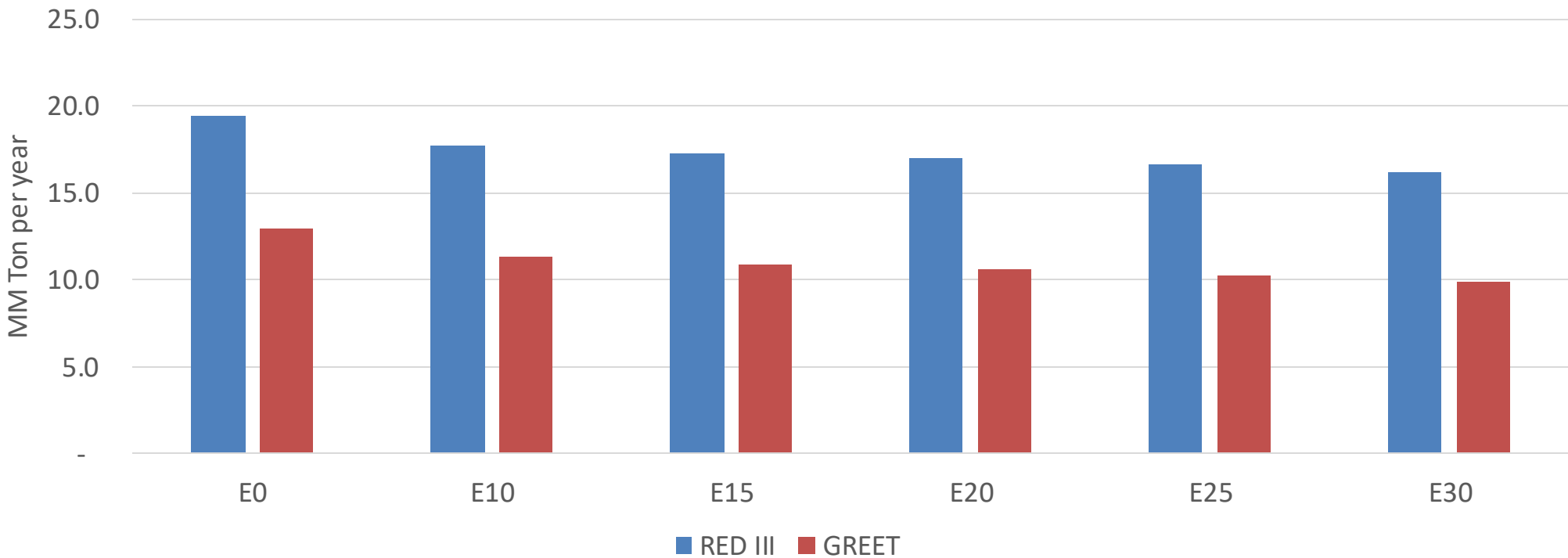
IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Greece – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2024 (MMT/año)	74.3
Target @ 2035 (MMT/year)	41.8
Delta	32.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.6%	15.7%	18.1%	20.5%	23.7%
RED II/III	0.0%	8.8%	11.0%	12.8%	14.5%	16.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.0%	6.2%	7.2%	8.2%	9.4%
RED II/III	0.0%	5.2%	6.6%	7.7%	8.7%	10.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90